

RE02 Real Estate Finance and Investment

Michaelmas Term 2026, Week 8

Urban and Regional Growth and Real Estate Pricing

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What this week does

The earlier weeks treat a market as a single place. This one asks why rents differ between one location and the next, and what happens to them when the region around them grows. Two frameworks do the work. Bid rent explains the pattern across space at a point in time, and a system of linked regional markets explains what happens to that pattern when demand, or population, or the capital stock changes. Blocks F and G then put both to the test against British and American evidence from the period since 2020, which is the largest shock to the geography of housing demand for which we have data.

A	Location and the structure of urban land markets	25 min
B	Regional growth and simultaneous equilibrium	40 min
C	Productivity, amenity and the correlation of wages and rents	20 min
D	Land rent and land leverage	30 min
E	Planning constraint and the implicit zoning tax	25 min
F	Housing, commercial property and the pandemic re-pricing	35 min
G	Measuring housing pressure before it reaches prices	30 min
H	Reserve: rent determination in retail	25 min

What you already have

Where this week draws on material taught elsewhere in RE02. Nothing below is assumed: each idea is stated in full here before the other lecture is named.

Where it was taught	What it gives this week	What happens if it was not covered
RE02 Week 2, space and capital markets	The four-quadrant model, and with it the result that any building can in principle be replicated at the cost of construction. Block E turns on that result.	Delivered by the same lecturer this year, so it can be relied on. Block E restates the replacement-cost argument in one sentence before using it.
The sector lecture on offices, retail and industrial, wherever it falls this year	Bid rent as the organising idea for urban land use, and the determination of retail rents.	Block A develops bid rent from first principles regardless of whether it has been seen. Block H holds the retail material in reserve in case the sector lecture did not reach it.
Any lecture on market efficiency, transparency or indices	The idea that transactions are infrequent and that measured prices adjust with a lag.	Nothing in this week depends on it. It is worth mentioning in Block D, where the land and structure split is estimated from repeat sales.

Block A

Location and the structure of urban land markets

Why the same building commands a different rent a mile away.

Indicative length 25 minutes

The question this week asks

Everything so far has treated a market as a single place with a single rent. That is a useful simplification and it is false in a way that matters, because the largest and most persistent differences in commercial rents are differences between locations rather than between periods. An office in the City and an identical office in an outer borough are the same building and not the same asset.

Two questions follow. What determines the pattern of rents across space at a point in time, and what happens to that pattern when the regional economy that supports it grows or contracts. Bid rent theory answers the first. A system of linked regional markets answers the second.

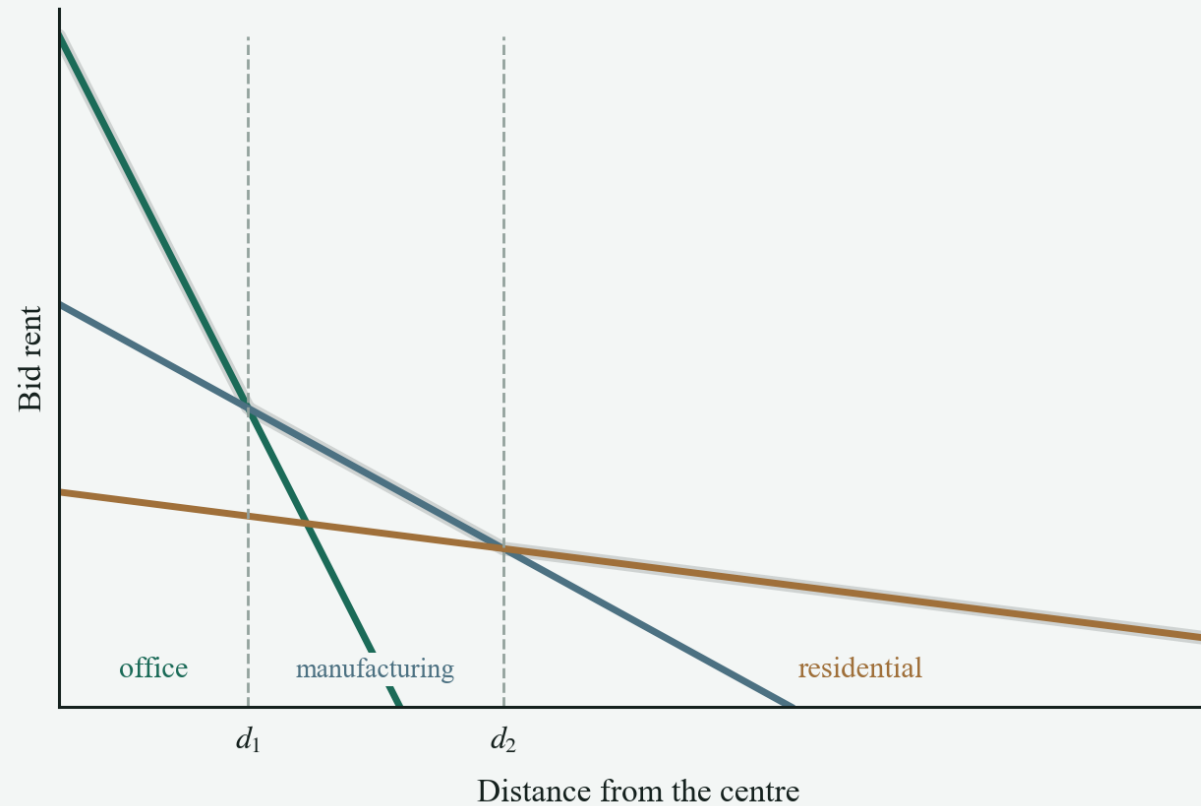
Bid rent from first principles

Consider a firm choosing a site. Wherever it locates it must reach its customers, its suppliers and its workers, and the cost of doing so rises with distance from the centre. If the firm is to be equally well off at any location it chooses, the rent it is willing to bid must fall by exactly the amount that its transport costs rise. That relationship, rent as a declining function of distance holding profit constant, is the bid rent curve.

The important step is the second one. Different activities have different sensitivities to distance. An occupier whose costs rise steeply with distance bids steeply, and therefore outbids everyone else at the centre and nobody at all further out. The pattern of land use we observe is the upper envelope of the individual bid rent curves, and the boundaries between uses fall where two curves cross.

The bid rent envelope and the pattern of land use

Three activities bid for the same sites. Office use has the steepest curve and wins the centre; residential use has the flattest and wins the periphery. The land use observed at any distance is whichever curve is highest there, so the envelope traced in grey is the rent actually paid.



What bid rent theory establishes

- Rent is a residual. It is what is left after an occupier has met every other cost of operating at a site, which is why rent absorbs locational advantage rather than causing it.
- The ordering of land uses from the centre outwards is determined by the steepness of the bid, and steepness reflects how costly distance is for that activity.
- Land use boundaries are where two curves cross, so a change in the transport cost of one activity moves boundaries rather than merely moving rents.
- The value of a site is the capitalised value of its locational advantage over the marginal site, which is the link back to the valuation quadrant of Week 2.
- Nothing in the argument requires anyone to behave strategically. The pattern emerges from competitive bidding alone.

What bid rent does not settle

The framework is static in the same way the four-quadrant model is static. It describes a configuration in which no occupier wishes to move, and it says nothing about how that configuration is reached, how long the adjustment takes, or what happens while it is under way. It also has no supply side worth the name: the quantity of land at each distance is given by geometry, and the possibility that planning might make some of it unavailable does not appear.

Both omissions matter for what follows. Block B supplies the missing account of adjustment by embedding the land market in a regional economy, and Block E supplies the missing supply constraint.

Block A discussion

Seminar questions

1. Remote working lowers the cost of distance for office employment. Which curve changes, in which direction, and what happens to the boundary between office and residential use?
2. Rent is described as a residual. If that is right, what does it imply about who bears the cost of an increase in business rates?
3. The envelope assumes competitive bidding for every site. Name a circumstance in which a landowner would rationally leave a site in a lower-value use.

Block B

Regional growth and simultaneous equilibrium

Three markets that have to clear at the same time.

Indicative length 35 minutes

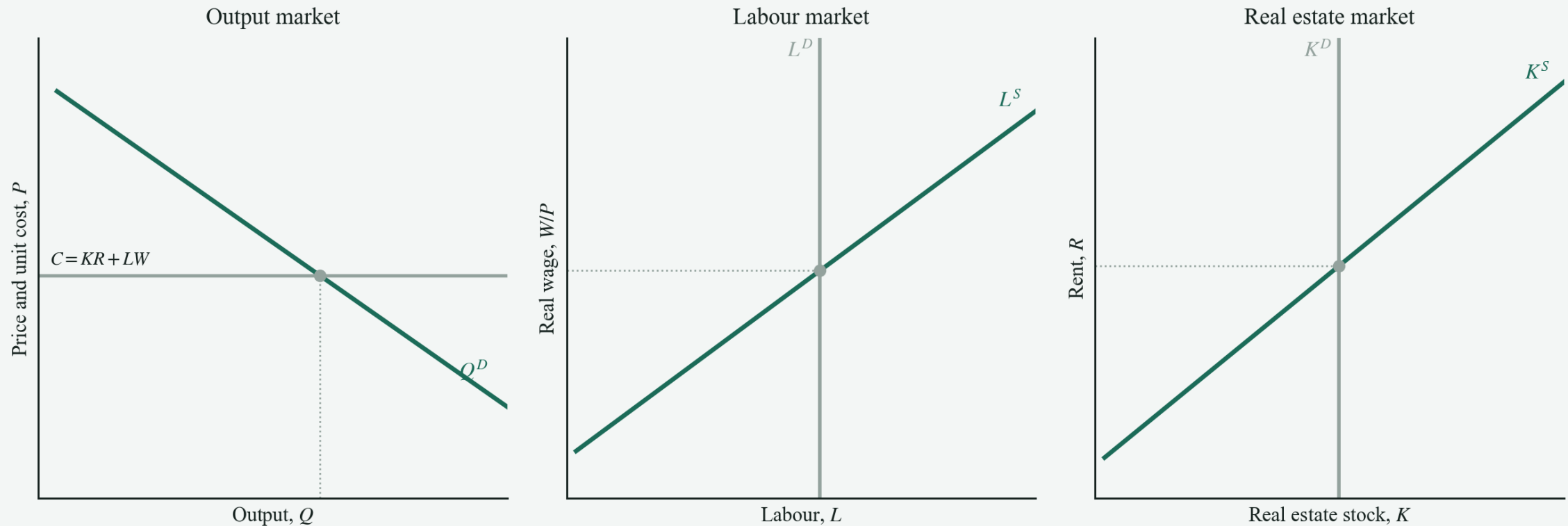
What determines demand in a region

- Export production, meaning goods and services sold outside the region. This is the classic base of regional growth and the part most exposed to national and international conditions.
- Local services, which are produced and consumed within the region and expand roughly in proportion to the population they serve.
- Government and private transfers, including social payments, pensions and investment income, which support demand without any corresponding local production.
- Direct investment, including inward investment in productive capacity and in the built stock itself.
- Meeting that demand requires factors of production. Labour comes from the resident population and from migration; capital, in this setting, is very largely real estate.

Three linked markets in simultaneous equilibrium

B

Output requires labour and space in fixed proportions, so the level of output fixes the demand for both. The wage and the rent that clear those two markets together determine unit cost, and unit cost has to equal price in the output market. The three panels are one system, not three diagrams.



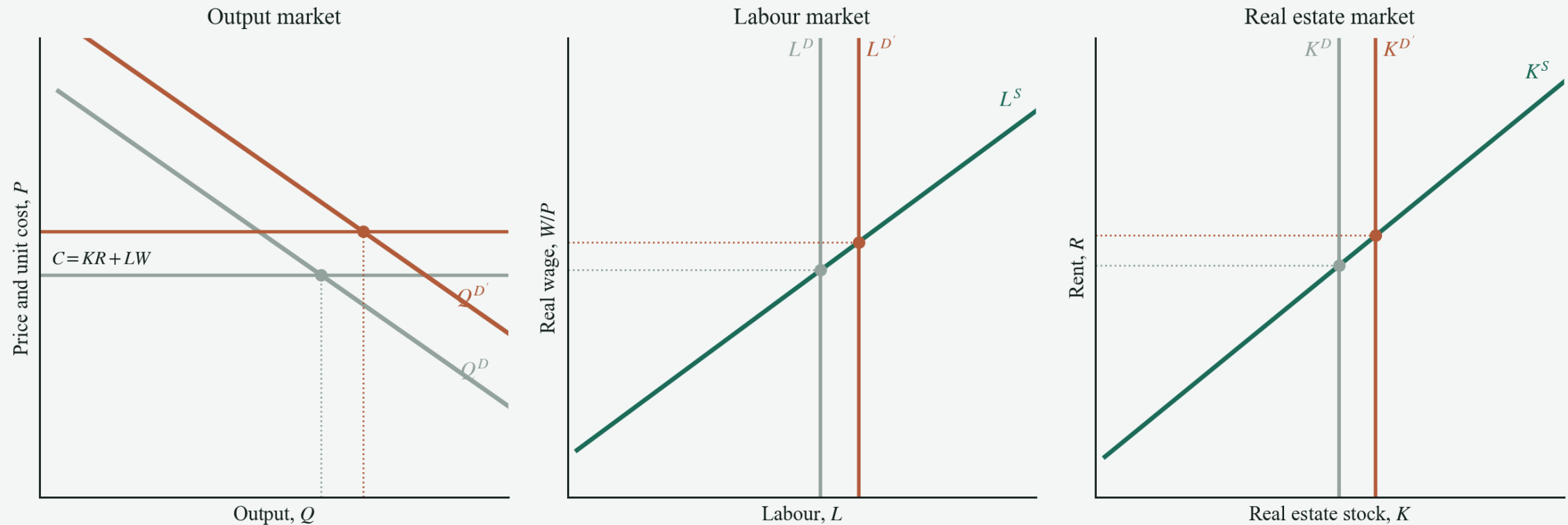
How the three markets are tied together

The chain has four links and it is worth stating each one. Demand for the region's output fixes how much is produced. Production requires labour and space in proportion to output, which fixes the position of the two demand schedules in the second and third panels. Those schedules meet upward-sloping supply schedules, which fixes the wage and the rent. The wage and the rent are the two components of unit cost, and unit cost determines the price at which the region's output can be sold.

The assumption doing the most work is that there is no substitution between labour and space. A firm facing an expensive location cannot economise by putting more workers in less floor area. That is plainly false, and relaxing it damps every result that follows without changing the direction of any of them.

An increase in demand for the region's output

Exports rise, so the demand schedule for output shifts out. Output expands, and with it the derived demand for labour and for space. Both wages and rents rise, unit cost rises with them, and the higher cost is what limits the expansion.

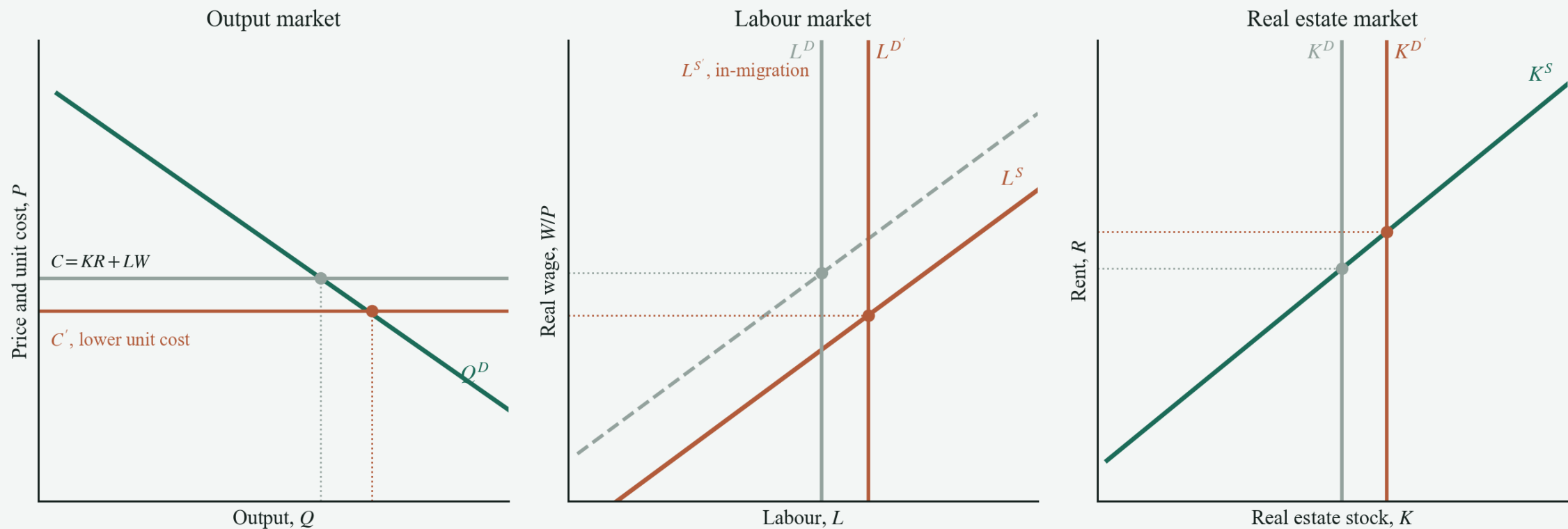


Reading the output demand shock

- Price and output both rise, which is the ordinary result of an outward shift in demand against an upward-sloping cost.
- Employment rises and, because the labour supply schedule slopes upward, so does the wage.
- The stock of space rises and so does rent, by the same argument applied to the third panel.
- A contraction runs the sequence in reverse, with the qualification that the stock of space cannot fall, so the adjustment shows up in vacancy and in price rather than in quantity.
- The elasticity of each supply schedule governs how the shock splits between quantity and price. A region with inelastic housing and commercial supply converts a demand shock almost entirely into higher rents.

An increase in the supply of labour

In-migration shifts the labour supply schedule out. The wage falls, and with it unit cost, so output expands. The expansion raises the derived demand for space, so rents rise even as wages fall.



Why in-migration lowers wages and raises rents

The two movements have the same cause and opposite signs, which is what makes the case instructive. More workers competing for jobs lowers the wage directly. The lower wage lowers unit cost, which allows the region to sell more output, and producing more output requires more space. The demand for space therefore rises even though the demand for labour per unit of output has not changed at all.

The result has an uncomfortable implication for the standard argument that a region should attract population in order to grow. It will grow, and the gain will accrue disproportionately to the owners of the fixed factor, which is land. Whether the incoming workers are better off depends on how much of their lower wage is absorbed by higher rent.

Does relocating economic activity actually move a local housing market?

The model says an exogenous increase in a region's output raises its rents. Places for Growth offers a rare chance to test that directly: the programme set out to move twenty-two thousand civil service roles out of London by 2030, and had moved some sixteen thousand into eight English hubs by late 2023. Each hub is a dated demand shock of known size.

The answer so far is that intent moved and cost did not. At Darlington, the most intensively treated hub at over twelve roles per thousand dwellings, a synthetic control finds no effect on rents or on prices, while inbound search for sales tilted towards southern origins by nearly five percentage points against comparison areas. Across all eight hubs a staggered difference-in-differences returns an aggregate rent effect of about five per cent with a clean pre-trend, which looks convincing until it is asked to scale: the dose-response is negative, so the hubs with the largest relocations show the smallest effects. What the design is picking up is most likely the post-2021 rent surge in large cities, not the relocations.

Block B discussion

Seminar questions

1. A region attracts a large inward investment in a single plant. Trace the effect through all three panels and say who gains most.
2. The model assumes no substitution between labour and space. Relax it. Which of the results in this block survive, and which are damped?
3. Output demand falls sharply and permanently. The stock of space cannot fall. What does the third panel look like a year later, and what variable is doing the adjusting?
4. The relocation result has a flat pre-trend and a negative dose-response. Which of the two would you weight more heavily, and what would settle it?

Block C

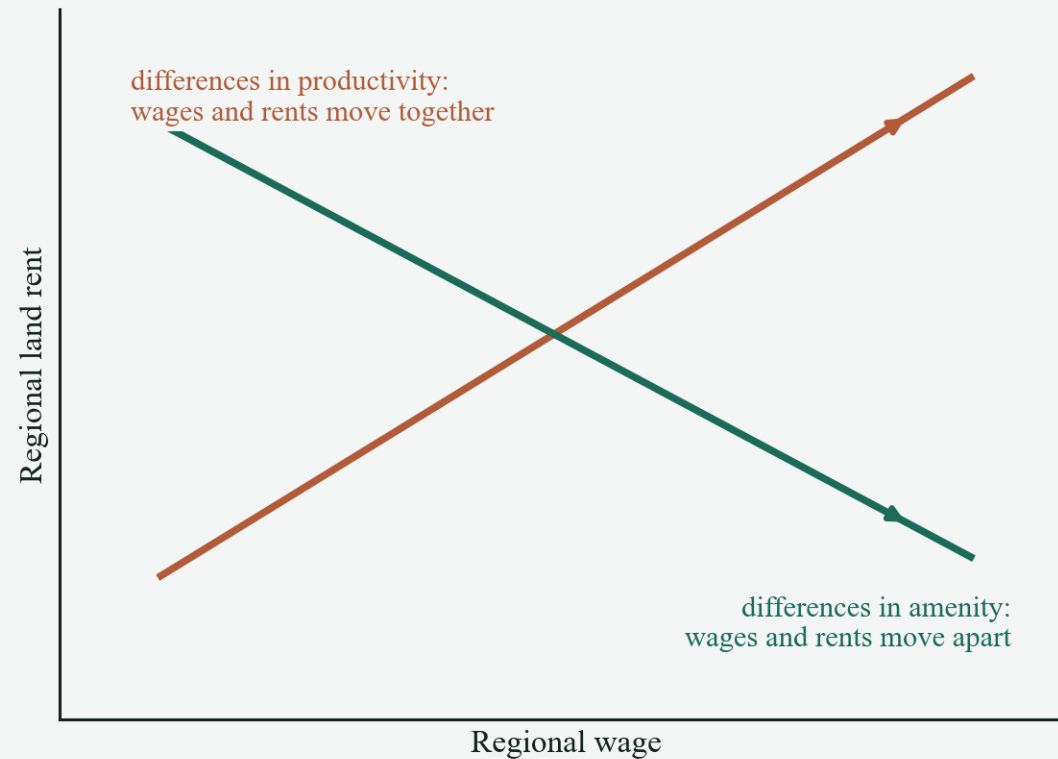
Productivity, amenity and the correlation of wages and rents

What the sign of a correlation reveals about why a place is expensive.

Indicative length 20 minutes

Why wages and rents sometimes move together and sometimes apart

Regions differ for two reasons, and the two leave opposite signatures. Where regions differ in productivity, the more productive place has both higher wages and higher rents. Where they differ in amenity, the more pleasant place has lower wages and higher rents, because workers accept less to live there.



Schematic, after the argument in diPasquale and Wheaton (1996), chapter 7.

Using the sign as a diagnostic

Suppose you observe that a city is expensive and you want to know why. The correlation across a set of comparable regions between the wage and the land rent tells you which mechanism dominates. A positive correlation points to productivity differences: firms can afford to pay more because output per worker is higher, and they bid more for space for the same reason. A negative correlation points to amenity: people will work for less in order to live there, and they will pay more for the privilege.

The investment implication is direct. Rents supported by productivity depend on the industries that generate it and will fall if those industries move. Rents supported by amenity depend on something that is usually fixed, which makes them more durable but also caps how far they can rise.

Where the diagnostic runs into trouble

- Amenity and productivity are correlated in practice. Places that are pleasant attract skilled workers, and skilled workers raise measured productivity.
- The wage that matters is the wage for a fixed quality of worker. Comparing raw averages across regions confounds the return to skill with the return to place.
- Land rent is rarely observed directly. What is observed is property value, which mixes land with structure, and Block D is about separating the two.
- Both mechanisms operate on the demand side. A region can also be expensive because supply is constrained, which the diagram does not represent at all and Block E takes up.
- The comparison requires a set of regions that are otherwise similar, and the choice of that set does much of the work in any published estimate.
- The whole exercise assumes workers can move to where the returns are. Where housing costs prevent that, the observed pattern of wages and rents reflects the constraint as much as the underlying advantage, which is the spatial misallocation argument of Hsieh and Moretti (2019) and of Monte, Redding and Rossi-Hansberg (2018).

Block C discussion

Seminar questions

1. Two cities have the same average wage and very different rents. What are the candidate explanations, and what further data would separate them?
2. If amenity-driven rents are more durable than productivity-driven rents, should an investor pay a lower yield for the former? Argue both sides.
3. Remote working weakens the link between where a worker lives and where the output is produced. What does that do to the diagnostic?

Block D

Land rent and land leverage

Separating the value of the site from the value of what stands on it.

Indicative length 30 minutes

Why land has to be separated from structure

A property is a bundle of two assets with almost nothing in common. The structure is reproducible, depreciates, and is worth roughly what it costs to build. The land underneath is not reproducible, does not depreciate, and is worth whatever the locational advantage established in Block A will support. They are bought together and priced together, and their returns behave quite differently.

Everything in this lecture so far has been about the land component. Bid rent determines the value of a site; the regional system determines what happens to that value when the region grows. None of it says anything about the building. Any empirical test of the argument therefore has to prise the two apart, and that is harder than it sounds, because the two are almost never transacted separately.

How Davis and Heathcote separate the two

- The value of a property is the value of the structure plus the value of the land, which is an identity rather than a finding.
- The structure component can be estimated from the outside, using a construction cost index and an estimate of the physical quantity of structure in place.
- The total value can be tracked using a repeat-sales index, which follows properties that transact more than once and so holds the bundle of characteristics fixed.
- Land value is then the residual: total value less the estimated structure value. The land price index is whatever is left after the construction cost of the building has been removed.
- The share of value accounted for by land follows directly, and it is the quantity that matters for everything that follows.

Land leverage, and why the same land price change is not the same value change

Two properties, identical in every respect except the share of value accounted for by land, face the same forty per cent rise in land prices. The property with the higher land share sees nearly three times the increase in total value.



Illustration constructed from the identity that value is land plus structure. No data source.

What land leverage explains

The land share behaves like an operating leverage. Because the structure component is anchored to construction cost and moves slowly, all of the volatility in land prices passes through to total value amplified by the reciprocal of the land share. Two properties in the same market, with the same tenants and the same lease terms, will therefore show quite different price volatility if their land shares differ.

Bostic, Longhofer and Redfearn use this to decompose house price dynamics, and Bourassa and colleagues show that the land share predicts the cross-section of price changes within a market. The practical reading is that land leverage is a risk factor which is not on most risk registers, and that a portfolio can be concentrated in it without anyone having chosen to be.

What you would need to replicate this

- A price index for gross investment in new residential structures, of the kind produced within national accounts, to value the structure component.
- A repeat-sales index for the total value of the bundle, so that changes in composition do not contaminate the series.
- An estimate of the replacement cost of the standing stock, usually built up through a perpetual inventory calculation from past investment less depreciation.
- A periodic benchmark for the physical stock, historically the decennial census of housing in the United States work, to anchor the inventory.
- For the United Kingdom the equivalents exist but are less well aligned, which is the main reason the exercise has been repeated less often here.

Block D discussion

Seminar questions

1. Land value is estimated as a residual. Construct a case in which an error in the construction cost index produces a spurious finding about land.
2. Two office buildings in the same city have land shares of thirty and seventy per cent. Which would you rather own going into a downturn, and what does your answer assume?
3. Does the land and structure distinction make sense for a property whose value is dominated by an existing tenant on a long lease?

Block E

Planning constraint and the implicit zoning tax

What it means when a building is worth more than it costs to build.

Indicative length 20 minutes

Replacement cost as the anchor

The construction quadrant of the four-quadrant model carries an implication that is easy to pass over. If any building can be replicated at the cost of construction, then no building can trade for much more than the cost of the land beneath it plus the cost of putting up the structure. Should it do so, someone will build another one and compete the difference away.

The argument only holds if building is actually possible. Where it is not, because permission will not be granted or because the permitted density is below what the market would support, the gap between what a building is worth and what it would cost to reproduce can persist indefinitely. That gap is the observable signature of a binding constraint on supply.

The implicit zoning tax

Glaeser and Gyourko turn that gap into a measure. Compare the market price of a property with the sum of the cost of the land it occupies and the cost of constructing the building on it. In an unconstrained market the two should be close. Where price exceeds cost persistently, the excess is the value of the permission to build, and it accrues to whoever holds the site rather than to anyone who produced anything.

They call the excess an implicit tax because it functions like one, raising the price paid by the occupier without generating revenue for the public purse. Its incidence is the interesting part: it is paid by occupiers and received by landowners, which makes constraint on new supply a transfer from the young and mobile to the old and settled.

What the measure captures, and what it does not

- It captures the value of the right to build at the observed density, aggregated over whatever mixture of constraints produces it, without attributing it to any particular rule.
- It requires a credible cost of land in an alternative use, which in a dense city is precisely the quantity that is hardest to observe.
- It cannot distinguish a constraint that is inefficient from one that is doing something worth doing. Amenity, heritage and the external effects of density are all real, and a constraint that protects them is not simply a tax.
- It is silent about who imposed the constraint. Physical limits, infrastructure capacity and the planning system all show up in the same number.
- The estimate is highly sensitive to the construction cost assumption, in the same way that the land residual in Block D is.

Why supply elasticity is the lever rather than one variable among many

Every result in this lecture has turned on the same parameter. The size of a rent spike after a demand shock, the share of a regional expansion that accrues to landowners rather than to workers, and the gap between price and replacement cost are all governed by how readily new supply appears. Saiz shows that the elasticity can be built up from geography and regulation together; Hilber and Vermeulen, working on English data, find the regulatory component the dominant one, which matters because geography cannot be legislated and regulation can.

That is what makes elasticity a policy lever rather than a nuisance parameter. Current departmental work takes the argument a step further and treats remote working as a labour-market shock whose efficiency gains and distributional harms both run through housing, with local supply elasticity determining how the two divide. The efficiency gain accrues to those who can move; the cost falls on incumbent renters in receiving areas who cannot work remotely. Where supply responds, more of the gain survives as broad welfare and less of it becomes a transfer.

Block E discussion

Seminar questions

1. A city has an implicit zoning tax of forty per cent of price. Name three explanations consistent with that observation, only one of which is a planning failure.
2. The measure needs the cost of land in an alternative use. How would you estimate it in central Cambridge, and what would go wrong?
3. If the constraint is a transfer from occupiers to landowners, what would a tax designed to recover it look like, and why have such taxes proved so hard to sustain?

Block F

Housing, commercial property and the pandemic re-pricing

The clearest natural experiment the field has had.

Indicative length 30 minutes

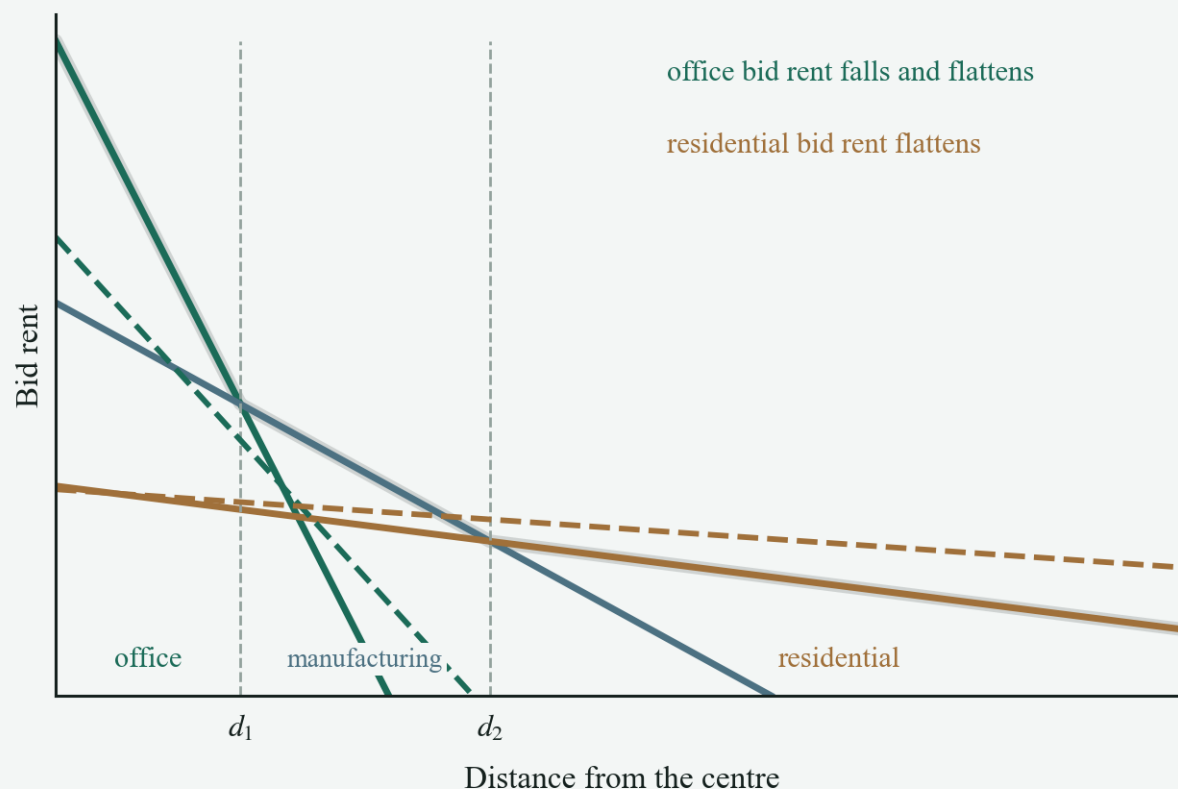
Complements or substitutes

Housing and commercial property are usually treated as separate asset classes with separate research, separate indices and separate investors. In a single urban land market they are neither separate nor independent: they compete for the same sites, and the boundary between them is exactly where their bid rent curves cross.

That makes them substitutes in the market for land and complements in the market for location. A district that becomes more attractive to residents becomes more attractive to the offices and shops that serve them, and the two bid against each other for the sites that result. Which effect dominates is an empirical question, and until 2020 it was very difficult to answer, because nothing moved the two curves independently.

A shock that moves the office and residential curves in different ways

Remote working reduces the number of days a worker travels to the centre. The cost of distance falls for both the firm and the household, so both curves flatten, and the office curve falls as well because less space is required per worker.



Schematic, after Alonso (1964); the empirical counterpart is in Gupta, Mittal, Peeters and Van Nieuwerburgh (2022).

What Gupta and colleagues found

The paper estimates the slope of the relationship between price, or rent, and distance from the centre, for postcode-level data across the largest metropolitan areas in the United States, and tracks how that slope moved through 2020. The finding is a flattening: prices and rents fell in the centre relative to the periphery, and the gradient that had been in place for decades became measurably shallower.

Two details matter more than the headline. The flattening was much larger for rents than for prices, which says that the market expected the change to be partly temporary and priced accordingly. And it was concentrated in the largest and densest cities, which is what the bid rent argument predicts, since those are the places where the cost of distance was highest to begin with.

What moved, and on which margin

The American evidence is about outcomes: prices and rents at different distances from the centre. It cannot say whether households were searching farther afield or simply whether more of them were searching beyond the boundary at all. That distinction matters, because the two imply different things about how durable the change will be.

British search data separates them. Work in this department using around 294 million property enquiries finds that the 2020 and 2021 shift was almost entirely an extensive-margin move: more households crossed their city boundary, while those who left did not search noticeably farther than leavers had before. The intensive margin was flat. The popular reading, that distance had ceased to matter, does not survive the decomposition, and over the longer run search distances had in fact been falling. The shift also diverged by tenure, reverting for buyers and persisting for renters.

Why rents moved more than prices

- A rent is a price for one year of occupation. A capital value is a price for the whole future stream, so a shock believed to be temporary moves the first far more than the second.
- The gap between the two movements is therefore an estimate of how permanent the market expected the change to be, which is a rare thing to be able to observe.
- Leases in place damped the measured movement in commercial rents, since existing contracts continued at pre-pandemic terms for years.
- Vacancy absorbed part of the adjustment in the office market rather than rent, which is the omission from the four-quadrant model catching up with us again.
- The residential and commercial markets adjusted at different speeds for the same reason: residential leases are short and commercial leases are long.

Conversion as the supply-side answer, and its limits

If the office core has contracted and the residential boundary has moved outward, the obvious response is to convert the surplus offices into housing. The Brookings work on this is a useful corrective, because it sets out how much of the standing office stock is actually convertible, and the answer is a small minority. Floor plates are too deep for residential daylight requirements, service cores are in the wrong places, and the structural grid rarely suits a residential layout.

The economics are worse than the engineering. Conversion is only viable where the office value has fallen far enough to make the residual land value under a residential scheme the higher of the two, and in most cities the office value has not fallen that far. The buildings that are easiest to convert tend to be the ones whose office value has held up.

Block F discussion

Seminar questions

1. Rents fell more than prices. Convert that difference into a statement about what the market believed in 2020, and say what would have had to be true for the reverse pattern.
2. The flattening was concentrated in the densest cities. Is that evidence for the bid rent framework, or is there an alternative explanation?
3. A city wants to convert its surplus offices to housing. Using Block A and Block E, what would have to be true for that to happen without a subsidy?

Block G

Measuring housing pressure before it reaches prices

Current work in this department, and what it does to the frameworks above.

Indicative length 30 minutes

Why prices are the wrong primitive

Everything in this lecture so far has been read off prices or rents. That is the standard practice and it has three known faults. Prices lag, because they capitalise demand slowly and only when a transaction happens. They reflect the composition of the stock that happened to trade, not the pressure on the market as a whole. And they are silent about where demand came from, which is exactly the question Blocks A to C are asking.

An alternative is to measure the imbalance directly, before it clears. Labour economics has done this for decades with the ratio of vacancies to unemployment. The housing analogue is the ratio of search demand to listed supply, and until recently there was no data with which to build it at national scale.

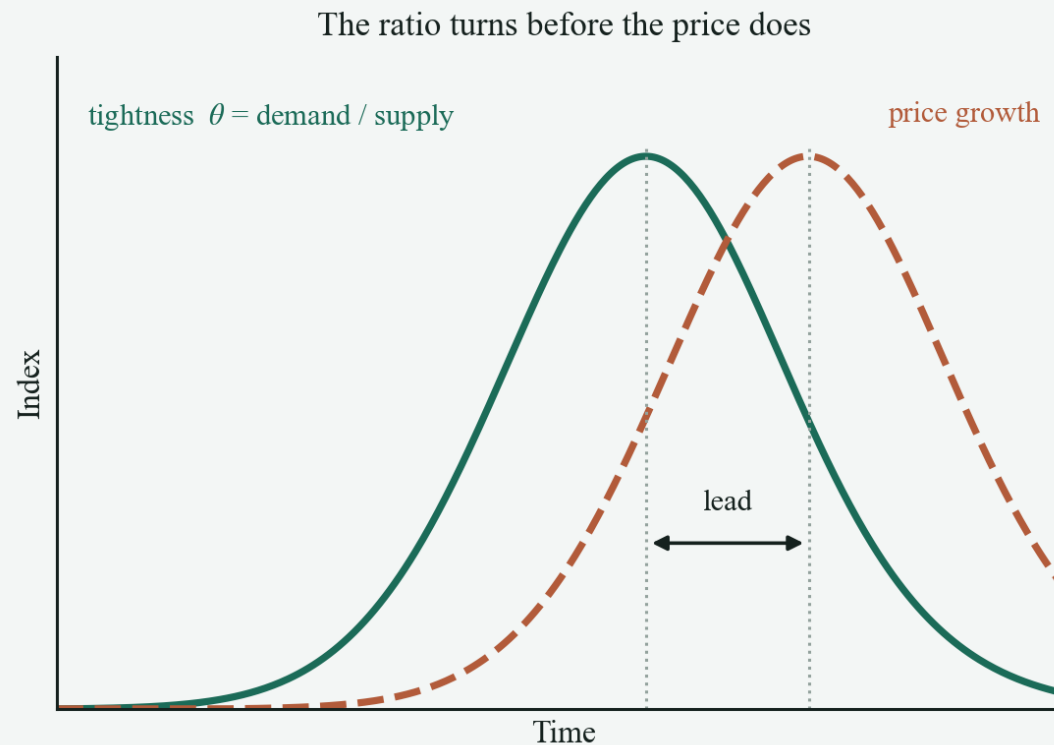
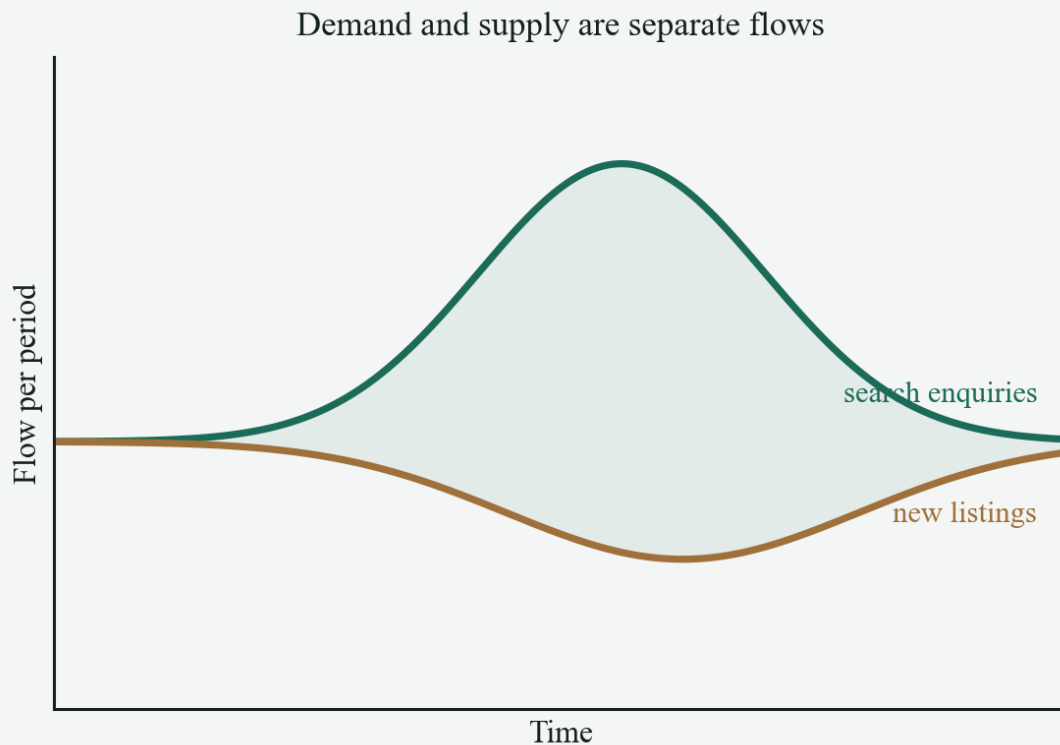
The data behind the measure

- Around 294 million property search enquiries for Great Britain from 2010 to 2025, each pairing a searcher's home location with the location of a property they are considering.
- The near-universe of listings over the same period, carrying asking prices and, where they match, Land Registry sold prices.
- Aggregated to local authorities and then to travel-to-work areas, giving a panel of roughly 150 areas by year, monthly where needed, for sales and rentals separately.
- What makes it unusual is the origin. Transactions, listings and census migration tables all record where demand landed; almost nothing records where it came from before any move occurred.
- The obvious caveats apply and bound everything that follows: coverage of resolved origins rises over the decade, so trends are read from 2016; and both demand and supply are scale measures, so it is the distribution of the ratio that is informative, not its level.

Tightness, and why a ratio of flows moves first



Search enquiries and new listings are separate flows and need not move together. Their ratio registers an imbalance as it forms, which is earlier than the point at which the imbalance clears through price.



Schematic. The shapes illustrate the construction and are not drawn from the estimated series.

What the tightness index shows for Britain, 2010 to 2025

Finding	Measure	Direction
Inequality of pressure across areas compressed under the pandemic, then diverged sharply	Theil index of sale tightness	0.058, falling to 0.048 in 2021, then rising to 0.095 by 2025
Rental pressure inequality did not compress at all	Theil index of rental tightness	Rising persistently, to 0.098
Spatial clustering weakened and then re-intensified	Global Moran's I on log tightness	0.54, to 0.48 in 2021, to 0.62 in 2025
London and the South East turned from unremarkable into significantly slack	Local clusters, 2019 against 2025	Six areas became significant cold clusters; ten hot clusters persisted
The divergence is increasingly within regions rather than between them	Between-region share of the Theil index	Falling from 45 per cent in 2019 to 36 per cent in 2025

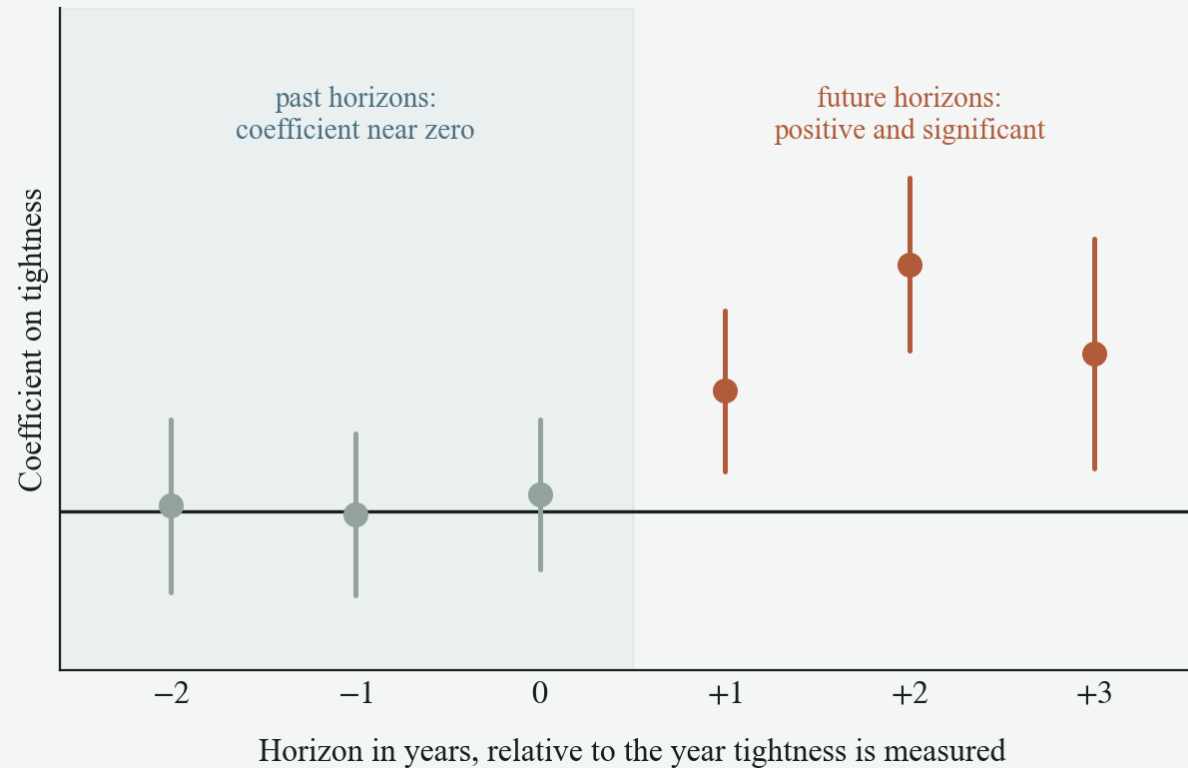
The pandemic dip was not what it looked like

The 2021 compression is easy to misread as a levelling of opportunity. It was the opposite: pressure was high almost everywhere at once, so the dispersion across areas fell while the level rose. Equality of pressure under a general shortage is not the same thing as equality of access, and an index that cannot tell them apart will mislead.

What followed is the more interesting result. By 2025 the dispersion had not merely recovered but exceeded anything in the preceding fifteen years, and the composition had changed: less of it is now accounted for by differences between regions and more by differences within them. On this evidence the flattening documented for American cities was a phase rather than a settlement.

The signature of a leading indicator

If a variable genuinely leads prices, its coefficient should be near zero when it is measured after the price movement and positive when it is measured before. A variable that merely moves with prices shows a symmetric pattern instead.



Schematic. The shape is the identification argument, not the estimated coefficients.

Search leads sales

The test is a fixed-effects panel of travel-to-work areas by year, regressing sold-price growth on tightness measured one and two years earlier, with area and year effects and standard errors clustered by area. Tightness predicts price growth one to two years ahead. The relationship survives a control for price momentum and in fact strengthens, which says it is forecasting something beyond the persistence prices already have. Run in reverse it disappears, so prices do not lead tightness. It replicates at local authority level.

Out of sample the gain is honest rather than spectacular. Adding tightness to an autoregressive benchmark improves forecasts around turning points, the reversal of 2020 and 2021 among them, and does little in ordinary years. That is the pattern one would expect of a pressure measure, and it is where a forecaster most needs help.

What this does not yet establish

Three limits are worth stating plainly, because they define the work that remains. The measures are descriptive. They document where pressure concentrated and show that it moved before prices, and they attribute none of it to any cause. The obvious candidate cause, remote working, is being tested separately, and that test needs data on which occupations can be done remotely and on how elastic local supply is, neither of which is in the search panel.

The welfare question is further off still. If remote working raised housing costs in receiving areas, the gain to those who moved and the loss to incumbent renters who could not are both real, and comparing them requires a model rather than a regression. That is the capstone of the programme and it is not yet written. Presenting it as a finding would be premature.

Block G discussion

Seminar questions

1. Tightness compressed in 2021 and diverged by 2025. Which of the two is the better description of a housing crisis, and what would you want to know before answering?
2. The between-region share of inequality is falling. What does that imply for a policy programme organised around closing gaps between the North and the South?
3. A leading indicator that helps only at turning points is of limited use unless you know a turning point is coming. Is it still worth having, and why?
4. What would have to be true for the flattening of the rent gradient to be permanent rather than a phase?

Block H

Reserve: rent determination in retail

Held in case the sector lecture did not reach it.

Indicative length 25 minutes

Why retail is treated separately

Retail rent is determined differently from office rent, and the difference is not a matter of degree. An office occupier bids for space on the basis of what it costs to house its workers. A retailer bids on the basis of what it expects to sell from that particular unit, and expected sales depend on the site in a way that expected office output does not.

The consequence is that retail rent is closer to a pure residual than any other kind. It is what remains from expected turnover after wages, stock and a required return on capital have been met, which makes it far more sensitive to location and far more responsive to trading conditions than a rent set by reference to a cost of accommodation.

What determines retail rents

- Supply is constrained in the long run by planning policy, which protects prime locations and restricts out-of-town development, so rents are demand-driven to an unusual degree.
- Proxies for that demand are retail sales, consumer expenditure, output and the cost of credit, and the correlation between consumer spending and retail rents is high.
- Micro-level determinants for shopping centres include age, size, design and the identity of the anchor tenant, which draws the footfall the other tenants rely on.
- Because desirable sites are few, competition for them takes the form of a bidding process, and the bid resolves into a residual payment.
- Forecasting retail rents therefore usually requires forecasting retail sales first, which imports every difficulty of consumer demand forecasting into the exercise.

Surplus theory: three retailers bidding for the same unit

	Firm 1	Firm 2	Firm 3
Expected turnover	100	120	110
Wages	30	30	30
Stock	50	50	50
Required return, 10 per cent	10	12	11
Surplus available for rent	10	28	19

Illustration. The surplus is turnover less wages, stock and the required return.

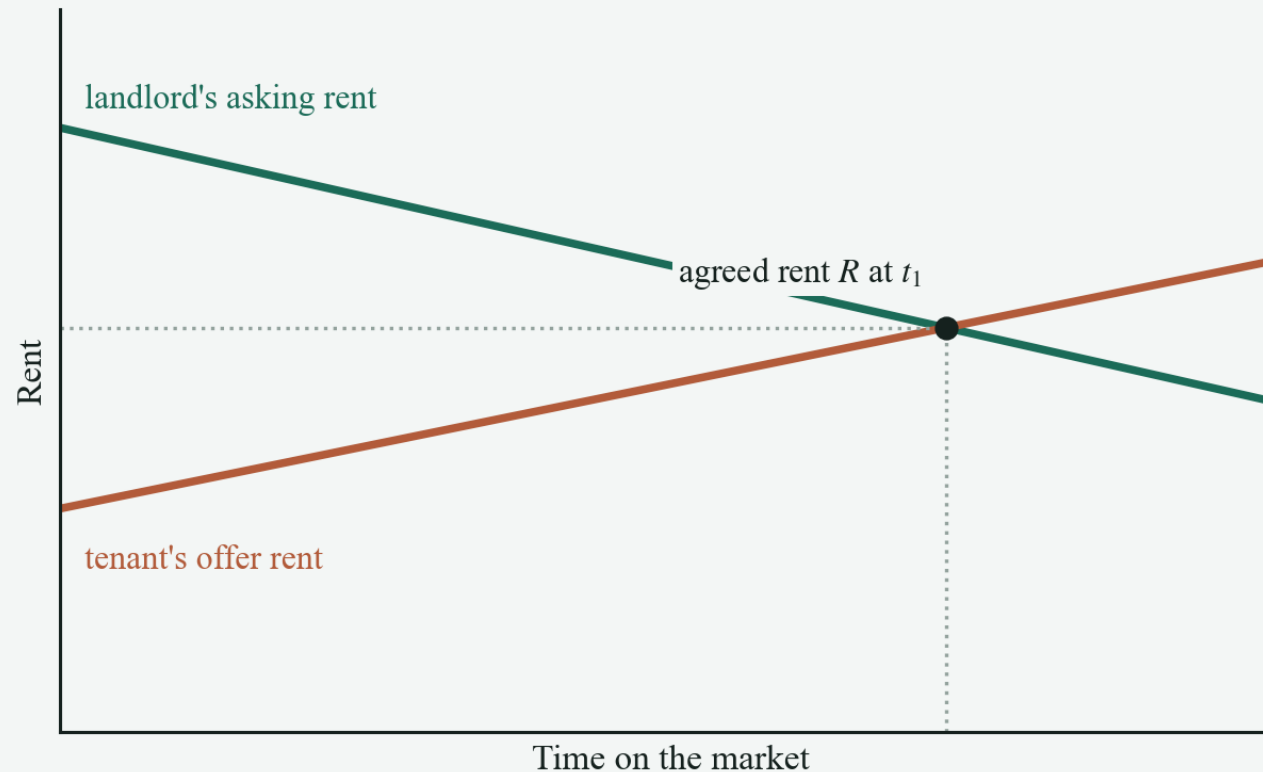
Percentage rent and the alignment of interests

A fixed rent leaves the landlord free to act against the interests of sitting tenants, by letting adjacent space to a competitor for instance, and to face no consequence until those tenants come to renew. A rent expressed as a percentage of turnover removes that freedom, because any action that harms a tenant's sales reduces the landlord's income immediately.

Wheaton makes this the central explanation for the practice, in preference to the risk-sharing account usually offered. The distinction matters because the two explanations predict different things: risk-sharing implies percentage rents should be commonest where turnover is most volatile, whereas the alignment argument implies they should be commonest where the landlord has the most scope to affect a tenant's trade, which is to say in managed centres.

Search and the time it takes a letting to happen

A landlord holding vacant space lowers the rent it will accept as the vacancy persists, and a tenant searching for space raises the rent it will offer as the search continues. The letting happens where the two meet, which fixes both the rent and the time on the market.



Schematic, after the search and matching literature; see Lisi (2015) for the application to the four-quadrant model.

How the frameworks in this module fit together

Four ideas have been used to explain the same thing, which is how a rent comes to be what it is, and each answers a different part of the question. Bid rent explains the pattern across space: which use wins which site, and why rent falls with distance. The four-quadrant model from Week 2 explains the relationship between that rent and the capital value built on it, and the supply response that follows. Search and matching explains the timing, which the other two leave out entirely. Surplus theory explains what happens where supply is so constrained that the site is effectively a monopoly, which is the retail case.

All four are inputs to a valuation rather than substitutes for one. The valuation lectures in Lent take the outputs of these frameworks as their assumptions, which is the practical reason for having spent a term on them.

Essential reading

- diPasquale, D. and Wheaton, W. (1996) Urban Economics and Real Estate Markets. Prentice Hall, chapter 7.
- Gupta, A., Mittal, V., Peeters, J. and Van Nieuwerburgh, S. (2022) Flattening the curve: pandemic-induced revaluation of urban real estate. Journal of Financial Economics 146(2), 594 to 636.

Further reading

- Alonso, W. (1964) *Location and Land Use: Toward a General Theory of Land Rent*. Cambridge, MA: Harvard University Press.
- Bostic, R. W., Longhofer, S. D. and Redfearn, C. L. (2007) Land leverage: decomposing home price dynamics. *Real Estate Economics* 35(2), 183 to 208.
- Bourassa, S. C., Hoesli, M., Scognamiglio, D. and Zhang, S. (2011) Land leverage and house prices. *Regional Science and Urban Economics* 41(2), 134 to 144.
- Davis, M. A. and Heathcote, J. (2007) The price and quantity of residential land in the United States. *Journal of Monetary Economics* 54(8), 2595 to 2620.
- Fingleton, B., Fuerst, F. and Szumilo, N. (2019) Housing affordability: is new local supply the key? *Environment and Planning A* 51(1), 25 to 50.
- Glaeser, E. L. and Gyourko, J. (2003) The impact of building restrictions on housing affordability. *Economic Policy Review* 9(2), 21 to 39.
- Hilber, C. A. L. and Vermeulen, W. (2016) The impact of supply constraints on house prices in England. *Economic Journal* 126(591), 358 to 405.
- Hsieh, C.-T. and Moretti, E. (2019) Housing constraints and spatial misallocation. *American Economic Journal: Macroeconomics* 11(2), 1 to 39.
- Loh, T. H., Terplan, E. and Rowlands, D. W. (2023) *Myths about converting offices into housing, and what can really revitalize downtowns*. Brookings Institution.
- Monte, F., Redding, S. J. and Rossi-Hansberg, E. (2018) Commuting, migration, and local employment elasticities. *American Economic Review* 108(12), 3855 to 3890.
- Plazzi, A., Torous, W. and Valkanov, R. (2010) Expected returns and expected growth in rents of commercial real estate. *Review of Financial Studies* 23, 3469 to 3519.
- Saiz, A. (2010) The geographic determinants of housing supply. *Quarterly Journal of Economics* 125(3), 1253 to 1296.

The Block G working papers

- Bao, H. X. H. (2026) Mapping the geography of housing demand: a family of search-flow indices for Great Britain, 2010 to 2025. Under review.
- Bao, H. X. H. (2026) Mapping spatial inequality in Britain's housing market: a demand and supply tightness index, 2010 to 2025. Working paper, in draft.
- Bao, H. X. H. (2026) Search leads sales: housing-market tightness as a leading indicator of prices. Working paper, in draft.
- Bao, H. X. H. (2026) Places for Growth and local housing markets. Working paper, in draft.
- The incidence and welfare papers referred to at the end of Block G are in design and have no results to cite.
- All four are provisional. Findings quoted in this lecture may change before publication, and students should cite them as working papers or ask before relying on them.

Where this leaves us

Rent falls with distance because the cost of reaching the centre rises with it, and the use that wins a site is the one whose bid falls most steeply. The pattern of land use is the upper envelope of the competing bids.

A region's output, labour and space markets clear together. A demand shock raises wages and rents together; an increase in labour supply lowers wages and raises rents. The sign of that correlation reveals whether a place is expensive because it is productive or because it is pleasant.

Land and structure behave differently and have to be separated before any of this can be tested. The land share governs how much of a land price movement reaches total value, and a gap between price and replacement cost measures the value of permission to build.

The pandemic flattened the rent gradient, and more for rents than for prices, which is as close to a measured expectation as this field gets. British search data adds that the move was on the extensive margin, and that the flattening was a phase: pressure inequality has since diverged past anything in the preceding fifteen years, increasingly within regions rather than between.